BTCPay Server: set up automatic installs of the Fiat on-ramp plugin

For the system administrator of the BTCPay Server host. Prepared 5 November 2026. Estimated time: 10 to 15 minutes, with one restart of BTCPay (a few seconds of downtime, choose a quiet moment).

What this is about

Our BTCPay Server gets a plugin called Fiat on-ramp. New versions of it will be installed automatically from GitHub Actions, so nobody has to upload files by hand after this. For that, the server needs a small install script and a dedicated SSH user whose key can run that script and nothing else.

You do not need a GitHub account. The plugin repository is public; the two files you download below are fetched anonymously. The GitHub side of the setup (secrets, variables) is done by Aviv.

What the script does is exactly what BTCPay's own "Upload plugin" button does: it copies the plugin package and its manifest into the plugins folder inside the btcpayserver container, queues an install command, restarts the container, and waits until BTCPay logs that the new version is running. Store settings live in BTCPay's database and are not touched.

What you will receive from Aviv: one line of text, the SSH public key (starts with ssh-ed25519). You need it in step 4.

What Aviv needs back from you: the server's hostname or IP and the SSH port, and confirmation that step 5 ended with OK.

Requirements

- BTCPay Server **2.4.5 or newer**, deployed the standard way (btcpay-setup.sh / docker-compose). The container is then named generated_btcpayserver_1 and the plugins folder inside it is /datadir/plugins.
- A user with sudo on the host, and docker available.
- Outbound HTTPS from the host to github.com (to download releases).
- Inbound SSH on some port reachable from the internet, because GitHub's build runners connect from changing IP addresses. If SSH is restricted to an allowlist, either add GitHub Actions' published IP ranges (api.github.com/meta, key "actions") or tell Aviv, and we switch to a variant where the server polls instead.

Quick checks:

```
docker ps --format '{{.Names}}' | grep btcpayserver
docker exec generated_btcpayserver_1 printenv BTCPAY_PLUGINDIR
command -v curl unzip
```

> Expected: the container name generated_btcpayserver_1; the plugin dir is usually empty or /datadir/plugins; curl and unzip present (install with apt install -y curl unzip if not). If the container has a different name, see Troubleshooting for the override.

Step by step

1. Create the deploy user

```
sudo useradd --create-home --shell /bin/bash btcpaydeploy
sudo usermod -aG docker btcpaydeploy
```

Membership of the docker group is what lets the script copy files into the container and restart it. The user gets no sudo rights.

If /etc/ssh/sshd_config has AllowUsers or AllowGroups, add btcpaydeploy there and run sudo systemctl reload ssh.

2. Install the script

```
sudo curl -fsSL -o /usr/local/bin/btcpay-install-plugin
https://raw.githubusercontent.com/mokokolalala/btcpayserver-plugin-fiatramp/master/deploy/
btcpay-install-plugin.sh
sudo chmod 755 /usr/local/bin/btcpay-install-plugin
```

Check that you got the exact file this document was written for:

```
sha256sum /usr/local/bin/btcpay-install-plugin
```

Expected checksum: `8f64725306df46689c152ecd43b79ee2a0718461f3cf65fececf63c652fdd065`

Feel free to read it (about 80 lines). It refuses any URL that is not a release of the plugin repository.

3. Prepare the SSH folder of the deploy user

```
sudo -u btcpaydeploy mkdir -p /home/btcpaydeploy/.ssh
sudo -u btcpaydeploy chmod 700 /home/btcpaydeploy/.ssh
```

4. Add Aviv's public key, locked to the script

Put the public key line you received into a variable, then write the authorized_keys entry. The command= prefix makes SSH ignore whatever the client asks to run and hand it to the script instead; the other options switch off port forwarding, agent forwarding and interactive shells for this key.

```
PUBKEY='ssh-ed25519 AAAA...paste the whole line you received here...'
echo "command=\"/usr/local/bin/btcpay-install-plugin\",no-port-forwarding,no-X11-
forwarding,no-agent-forwarding,no-pty $PUBKEY" | sudo -u btcpaydeploy tee -a
/home/btcpaydeploy/.ssh/authorized_keys >/dev/null
sudo -u btcpaydeploy chmod 600 /home/btcpaydeploy/.ssh/authorized_keys
```

Check the result; it must be one line, starting with command= and ending with the key and its comment:

```
sudo cat /home/btcpaydeploy/.ssh/authorized_keys
```

5. Test: install the current release (this restarts BTCPay)

You can test without the private key by running the script directly as the deploy user. This performs a real install of version 0.2.1:

```
sudo -u btcpaydeploy /usr/local/bin/btcpay-install-plugin
https://github.com/mokokolalala/btcpayserver-plugin-fiatramp/releases/download/v0.2.1/
BTCPayServer.Plugins.FiatRamp.btcpay
```

Expected output (last lines):

```
[install-plugin] restarting generated_btcpayserver_1
[install-plugin] waiting for BTCPay to load the plugin
info: BTCPayServer.Plugins.PluginManager: Running plugin BTCPayServer.Plugins.FiatRamp -
0.2.1.0
[install-plugin] OK
```

If it ends with OK, the setup is complete. In the BTCPay web interface, Server Settings > Plugins now lists "Fiat on-ramp 0.2.1".

6. Report back

Send Aviv: the hostname or IP to SSH to, the SSH port, and "step 5 OK". Aviv then enters these in GitHub, and the first automatic deploy happens with the next release. From then on, each release shows up in BTCPay a minute or two after it is tagged, with a short restart.

Afterwards: what runs on this server

- On each release, GitHub connects as btcpaydeploy and sends one argument: the release URL. The script downloads from github.com, installs, restarts BTCPay, and reports the result back into the GitHub job log.
- Nothing else can be run with that key. You can see every connection in /var/log/auth.log (user btcpaydeploy).
- To stop the automatic deploys at any time: remove the line from /home/btcpaydeploy/.ssh/authorized_keys. Removing the plugin itself: BTCPay > Server Settings > Plugins > Fiat on-ramp > Uninstall, then restart.
- Manual install or rollback, without GitHub: run the step 5 command with another release URL, replacing `v0.2.1` in both places with the wanted version. The release list is public at github.com/mokokolalala/btcpayserver-plugin-fiatramp/releases.

Troubleshooting

Symptom	What to do
container generated_btcpayserver_1 not found	Find the real name with docker ps, then run the script with BTCPAY_CONTAINER=<name> in front, e.g. sudo -u btcpaydeploy env BTCPAY_CONTAINER=btcpayserver /usr/local/bin/btcpay-install-plugin <url>. To make it permanent, put export BTCPAY_CONTAINER=<name> in /home/btcpaydeploy/.bashrc, or edit the default at the top of the script. Same for BTCPAY_PLUGINDIR if the plugins folder is not /datadir/plugins.
refusing URL outside ...	The script only accepts URLs of the form https://github.com/mokokolalala/btcpayserver-plugin-fiatramp/releases/download/<tag>/BTCPayServer.Plugins.FiatRamp.btcpay. Check for typos.
manifest ... not found / curl error	The host cannot reach github.com over HTTPS, or the release tag does not exist. Try curl -I https://github.com from the host.
plugin did not report as running within the timeout	BTCPay took longer than 3 minutes to start, or the plugin failed to load. Check docker logs --since 10m generated_btcpayserver_1 | grep -i plugin. If the plugin threw an error, BTCPay disables it and still starts; Server Settings > Plugins shows the reason. Report the lines to Aviv.
permission denied on docker	The deploy user is not in the docker group yet (new group membership applies to new sessions). Re-run with sudo -u btcpaydeploy, which starts a fresh session, or log out and in.
GitHub deploy fails with Permission denied (publickey)	The authorized_keys line is malformed, has the wrong permissions (must be 600, folder 700), or sshd restricts users. Compare with step 4; check /var/log/auth.log while a deploy runs.

Reference copies of the script and this procedure are kept in the public repository under `deploy/`. Questions: ask Aviv.